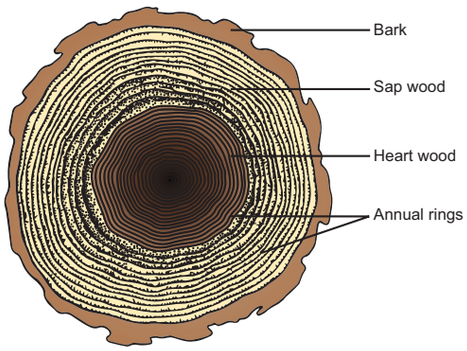


Figure 10.10: Cross - section of wood showing annual ring

Differences Between Sap Wood (alburnum) and Heart Wood (duramen)

Sap Wood (Alburnum)	Heart Wood (Duramen)
Living part of the wood.	Dead part of the wood.
It is situated on the outer side of wood	It is situated in the centre part of wood
It is pale coloured	It is dark coloured
Very soft in nature	Hard in nature
Tyloses are absent	Tyloses are present
It is not durable and not resistant to microorganisms	It is more durable and resists microorganisms

Secondary Phloem

The vascular cambial ring produces secondary phloem or bast on the outer side of the vascular bundle.

Just as the secondary xylem, the secondary phloem also has two tissue systems – the axial (vertical) and the radial (horizontal) systems derived respectively from the vertically elongated fusiform initials and horizontally elongated ray initials of vascular cambium. While sieve elements, phloem fibre, and phloem parenchyma represent the axial system, phloem rays represent the radial system. Life span of secondary phloem is less compared to secondary xylem. Secondary phloem is a living tissue that transports soluble organic compounds made during photosynthesis to various parts of plant.

Some commercially important phloem or bast fibres are obtained from the following plants.

i. Flax-*Linum usitatissimum*
ii. Hemp-*Cannabis sativa*
iii. Sun hemp-*Crotalaria juncea*
iv. Jute-*Corchorus capsularis*

Be friendly with your environment (Eco friendly)

Why should not we use the natural products which are made by plant fibres like rope, fancy bags, mobile pouch, mat and gunny bags etc., instead of using plastics or nylon?

Periderm

Whenever stems and roots increase in thickness by secondary growth, the periderm, a protective tissue of secondary origin replaces the epidermis and often primary cortex. The periderm consists of phellem, phellogen, and phelloderm.

Phellem (Cork)

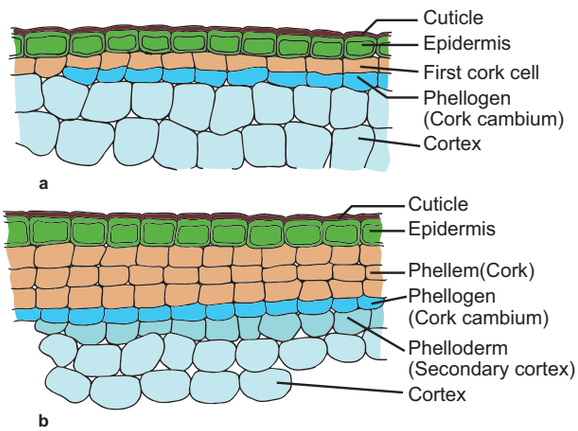


Figure 10.11: The cross section of periderm (a–b)

It is the protective tissue composed of non-living cells with suberized walls and formed centrifugally (outward) by the phellogen (cork cambium) as part of the periderm. It replaces the epidermis in older stems and roots of many seed plants. It is characterized by regularly

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